COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 - DATA STRUCTURES AND ALGORITHMS	**SECTION:**	CPE21S1
DATE PERFORMED:	Oct 6, 2026	**DATE SUBMITTED:**	Oct 6, 2026
NAME:	Corro, Yazinel Obias, Jave Joaquin	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

ACTIVITY NO. 11.1 [Basic Algorithm Analysis]

1. PROCEDURE OUTPUT

ILO A: Measure the runtime of algorithms using theoretical analysis

Given the above information, the total number of comparisons in the worst case is:

ANALYSIS:	
Corro	I looked at the graph and data table to see how the workload grows as the lists get larger. For list sizes of 10, the algorithm performs 110 operations, for list sizes of 20 the operations jump to 420 and at a size of 30 it reaches 930 operations. The graph shows a steep upward curve, meaning that the number of required operations grows much faster as the list size increases
Obias	Looking at the data table, the total operations go from 110 at size 10, to 420 at size 20, up to 930 at size 30. The graph's steep curve shows that larger lists require significantly more operations, meaning the algorithm slows down faster as the input size increases.

ILO B: Analyze the results of runtime performance by comparing experimental and theoretical analysis

Input Size(N)	Execution Speed	Screenshot	Observation(s)

ANALYSIS:	
Corro	
Obias	

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

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ILO A: Measure the runtime of algorithms using theoretical analysis & ILO B: Analyze the results of runtime performance by comparing experimental and theoretical analysis

Problem 1:

Problem 2:

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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I affirm that I will not give or receive any unauthorized help on this activity/exam and that all work will be my own.